

Mohammad Afzal

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Summary

Senior Staff Engineer / Lead Backend Engineer with **10+ years of experience** designing, modernizing, and scaling secure enterprise applications across **digital banking, insurance, fintech, SaaS, ERP, and product platforms**. Strong expertise in **Java, Spring Boot, microservices, distributed systems, AWS, Kubernetes, CI/CD, cloud-native architecture**, and production-grade system design. Experienced in **leading engineering teams**, owning end-to-end delivery, modernizing legacy applications, optimizing performance, resolving production issues, and integrating backend systems with frontend platforms, third-party services, and enterprise security solutions. Known for **strong ownership, release execution, banking-grade reliability, clean architecture, cloud automation**, and practical engineering leadership. Active Stack Overflow contributor with **8K+ reputation**.

Technical Proficiencies

- **Languages:** Java 21 / Java 17, Kotlin, Python, JavaScript
- **Backend Engineering:** Spring Boot, Hibernate/JPA, REST APIs, Microservices, Node.js, Kafka, RabbitMQ, API Design, Distributed Systems
- **Cloud & Platform:** AWS, EKS, ECS, EC2, Lambda, S3, RDS, API Gateway, EventBridge, SNS, SQS, Step Functions, IAM, CloudWatch, Azure
- **DevOps & CI/CD:** Docker, Kubernetes, Helm, Terraform, Jenkins, GitHub Actions, Linux CLI, Nginx
- **Databases & Caching:** PostgreSQL, Oracle, MySQL, MongoDB, DynamoDB, Redis, Snowflake
- **Frontend:** ReactJS, Angular, AngularJS, Flutter
- **Observability & Quality:** Datadog, New Relic, ELK, Codacy, JUnit, JMeter, Cucumber, Selenium
- **MLOps & Data Engineering:** Databricks, MLflow, SageMaker, Scikit-learn, JupyterLab, Pandas

Experience

Senior Staff Engineer, First Abu Dhabi Bank (via Nagarro) | Oct 2025 – Present

- Leading a team of 5 engineers in delivering and modernizing mission-critical **digital banking applications** for one of the UAE's largest banking platforms.
- Developed and enhanced secure backend services using **Java 21, Spring Boot, REST APIs, microservices**, and enterprise integration patterns.
- Built and supported AWS-based workloads using **EKS, API Gateway, EventBridge, SNS, SQS, DynamoDB**, along with Oracle and PostgreSQL for enterprise banking systems.
- Coordinated production releases, late-night deployments, hotfixes, and high-priority incident resolution to maintain continuity of banking operations.
- Collaborated with product owners, QA teams, frontend engineers, and external partners to deliver compliant, reliable, and production-ready banking features.

Lead Cloud Engineer, Liberty Mutual (via Technumen) | Mar 2024 – Jul 2025

- Designed and developed a **Spring Boot SDK with Spring WebFlux** to expose AI-powered insurance claim inferences through scalable enterprise API interfaces.
- Led development of the **OneRule SDK API layer**, enabling third-party systems to consume inference results with high reliability, low latency, and simplified integration.
- Built cloud-native deployment and automation workflows using **AWS EKS, Lambda, Step Functions, S3, CloudWatch, IAM**, and GitHub Actions.
- Created reusable platform assets including **Helm charts, EKS deployment templates**, and CI/CD workflows to accelerate onboarding and platform adoption.

- Integrated enterprise systems including **Microsoft Graph API**, **Slack bots**, **email pipelines**, **S3**, and **OpenAI APIs** for automation and internal workflow optimization.
- Contributed to architecture decisions, code reviews, developer mentoring, and engineering standards across cloud-native and AI-driven insurance platforms.
- Strengthened observability, code quality, and release confidence using **Datadog**, **Prometheus**, **Grafana**, **Sonar**, **Codacy**, and GitHub-based quality checks.

Product Development Engineer, Calibo | Oct 2020 – Jan 2024

- Lead the development of scalable web applications in a microservice architecture using Java, Spring Boot, and Hibernate.
- Collaborate with cross-functional teams to understand requirements and deliver high-quality features to production.
- Architect and implement DevOps practices, leveraging Docker, Kubernetes, and Jenkins for continuous integration and deployment.
- Utilize **AWS** services such as **EC2**, **RDS**, and **Sagemaker** to build robust and scalable cloud solutions.
- Work closely with stakeholders to gather feedback, optimize application performance, and ensure smooth product delivery.

Full Stack Developer - Northern Trust (via eTeams) | Sep 2023 – Jan 2024

- Played an integral role in building and maintaining a microservices architecture, including core components such as **Spring Cloud Config Server** and **Eureka Server**.
- Led security enhancements and implemented **Azure AD-based SSO** across multiple services, strengthening authentication and access control.
- Engineered **CI/CD pipelines using GitHub Actions**, improving development and deployment efficiency.
- Performed comprehensive unit and performance testing using **JUnit** and **JMeter**, ensuring system reliability and efficiency.
- Utilized **BDD with Cucumber** to align software development with business requirements and acceptance criteria.

Product Development Engineer, Calibo | Oct 2020 – Sep 2023

- Led the development of **scalable microservices** using **Java (Spring Boot + Hibernate)** and **Node.js**.
- Built and integrated data pipelines with **Snowflake** to support ETL workflows for ML Models.
- Architected and implemented DevOps practices using **Docker**, **Kubernetes (EKS)** with **Helm Charts**, **Jenkins**, and **Terraform** to manage infrastructure as code and automate environment provisioning.
- Built cloud-native applications leveraging **AWS services** including **EKS**, **ECS**, **RDS**, and **SageMaker**.
- Implemented GitOps-driven delivery pipelines using **Argo CD** to manage Kubernetes resources declaratively, supporting faster and more reliable platform releases.

Senior Associate, Nagarro | Mar 2019 – Oct 2020

- Developed enterprise applications using a **hybrid microservices stack**—**Node.js** for lightweight APIs and **Kotlin and Java with Spring Boot** on **AWS (Lambda, EKS etc)**.
- Led a small team of engineers in designing, developing, and maintaining microservices.
- Designed and developed scalable web applications using **Kotlin**, **Spring Boot**, **RabbitMQ**, **Angular 8** with **typescript**, and **PostgreSQL**.
- Actively participated in agile development processes, including sprint planning, daily stand-ups, and retrospective meetings.

Senior Software Engineer, Enbraun | Jul 2015 – May 2019

- **Core contributor and technical lead** on the development of *eResource Scheduler*, a multi-user resource management platform built from the ground up for scalability, availability, and customization on **AWS (EC2, RDS, S3, ALB)**.
- Designed and developed **RESTful services** and dynamic frontend modules using **JavaEE**, **AngularJS**, and **Bootstrap**, forming the backbone of the platform's booking, timesheet, and reporting features.
- Mentored junior developers and facilitated code reviews, fostering a culture of quality and hands-on learning within the team.

- Played a core role in the platform's **first public release after 3 years of active development**, contributing across all phases—from designing architecture and building features to testing, optimization, and deployment of a **highly available and scalable system**.

Project Profile

Liberty Mutual - OneRule SDK - AI Inference Platform for Insurance Claims

An enterprise-grade SDK built to simplify and standardize the use of Sapient's AI-driven insurance claim inference engine. The SDK acted as a middleware abstraction layer, enabling client systems to integrate seamlessly without dealing with data preparation, complex dependencies, or cloud-specific configurations.

Key Contributions:

- **Designed and built the core OneRule SDK** using **Java 17 + Spring Boot**, exposing an intuitive API layer for secure and scalable AI inference integration.
- Developed reusable infrastructure kits, including **Spring Boot Starters**, **Spring Webflux**, **Helm charts**, and **EKS-ready deployment templates** to support rapid onboarding and consistent rollout.
- Built **cloud-native automation pipelines** using **EKS**, **AWS Lambda**, **Step Functions**, **S3**, **CloudWatch**.
- Built a voice-driven Customer Care Assistant Bot for IVR System with **LangChain**, **GPT API**, **Whisper ASR**, and **Coqui TTS** for real-time conversational AI in Python.
- Migrated repositories from Bitbucket to GitHub, and implemented **GitHub Actions** for CI/CD, integrated with **Codacy** and **Datadog** for real-time observability and alerting.
- Integrated enterprise tools such as **Microsoft Graph API** (email/calendar automation), **Slack bots**, **OpenAI APIs**, and **custom email pipelines**, enhancing internal workflows and communication automation.

LAZSA - Product Platform-as-a-Service

Industry-first Product Platform-as-a-Service (pPaaS) to redefine digital innovation and drive competitive advantage through digital business. The platform provides a comprehensive set of tools and technology stack to efficiently manage, define, design, develop, and deploy phases of product development.

Link : <https://lazsa.altimetrik.com/>

Key Contributions:

- **Led the engineering team** in building and delivering core platform capabilities.
- Designed and implemented microservices using **Kotlin and Java with Spring Boot**, **Python** and **Node.js**, selecting the best-fit tech based on functional and performance requirements.
- Built a secure and scalable **DevSecOps pipeline** using **Docker**, **Kubernetes**, **EKS with Helm Charts**, **Jenkins**, and **Terraform** for infrastructure as code (IaC)
- Integrated advanced tools like **Kafka**, **SageMaker**, **MLflow**, **Kedro**, and **ELK Stack** to support real-time data processing, machine learning workflows, and observability.
- Designed the data and experimentation layer using **Snowflake** for ETL workflows and **JupyterLab** for EDA, ML experimentation, and interactive reporting.

DiPEC (Digitalized Project Execution Cycle)

Web application created to digitalize project development, utilized by engineers of Siemens. The project involved team leadership and development from the initial stages. The application was developed in a microservices architecture, ensuring scalability.

Link : <https://www.dipec.siemens.co.in>

Key Contributions:

- Took ownership of feature development from design to deployment, building scalable microservices and intuitive frontends based on Siemens' engineering needs.
- Designed and implemented services using **Spring Boot**, **PostgreSQL** and **Kafka** on **AWS EKS**.
- Developed a dynamic and responsive frontend using **Angular 8** and **Material UI**.
- I was involved in code quality, reviews, and sprint delivery in agile processes.

eResource Scheduler

Enbraun's eResource Scheduler is a multi-user feature-rich software for resource management and scheduling. The application enables organizations to efficiently schedule, plan, and manage employees, equipment, and other resources.

Link : <https://www.eresourcescheduler.com>

Key Contributions:

- Developed scalable **RESTful APIs** and interactive frontend modules using **JavaEE**, **AngularJS**, **Bootstrap**, and **OAuth2** on **AWS (EC2, RDS, S3 etc)**.
- Implemented advanced features including **user-defined fields**, **dynamic booking charts**, **timesheet modules**, and **customized reporting**, significantly improving planning flexibility and user experience.

ExamChi – Online Exam Practice Platform

A cloud-based exam preparation tool designed to deliver dynamic quizzes, performance tracking, and practice sessions for students and professionals. Built as a full-stack application to demonstrate scalable architecture, real-time user interaction, and cloud-native deployment.

Link : <https://examchi.afzalex.com/>

Key Contributions:

- Designed and developed the platform using **Node.js** (Express) for backend services, **ReactJS** for the frontend, and **MongoDB** for schema-flexible data storage.
- Deployed on **AWS** with a focus on performance and availability, leveraging services like EC2 and S3 for hosting and storage.

Education Detail

Master of Science in Data Science from Liverpool John Moores University (Distance)

Post Graduate Diploma in Data Science from IIITB Bangalore (Distance)

Bachelor of Technology in Computer Science from Uttarakhand Technical University